

What Is the Correlation Coefficient?

Correlation is a statistical measure that describes the strength and direction of a relationship between two variables.

The correlation coefficient is a statistical measure of the strength of a linear relationship between two variables.

Its values can range from -1 to 1.

A correlation coefficient of -1 describes a perfect negative, or inverse, correlation, with values in one series rising as those in the other decline, and vice versa.

A coefficient of 1 shows a perfect positive correlation, or a direct relationship.

A correlation coefficient of 0 means there is no linear relationship.

Key Characteristics of Correlation

1. Direction:

- **Positive Correlation:** As one variable increases, the other also increases.

Example: Height and weight often show a positive correlation.

- **Negative Correlation:** As one variable increases, the other decreases.

Example: The more a car is driven, the less fuel remains in the tank.

- **No Correlation:** No apparent relationship between the variables.

Example: Shoe size and intelligence.

2. Strength:

- The strength of correlation is expressed as a value between **-1 and +1**, called the **correlation coefficient**.
 - **+1:** Perfect positive correlation.
 - **0:** No correlation.
 - **-1:** Perfect negative correlation.

Pearson Correlation Coefficient (r):

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Values of r:

- $r > 0$: Positive correlation.
- $r < 0$: Negative correlation.
- $r = 0$: No correlation.

Pearson correlation coefficient (r) value	Strength	Direction
Greater than .5	Strong	Positive
Between .3 and .5	Moderate	Positive
Between 0 and .3	Weak	Positive
0	None	None
Between 0 and -.3	Weak	Negative
Between -.3 and -.5	Moderate	Negative
Less than -.5	Strong	Negative

Visualizing the Pearson correlation coefficient

Another way to think of the Pearson correlation coefficient (r) is as a measure of how close the observations are to a line of best fit.

The Pearson correlation coefficient also tells you whether the slope of the line of best fit is negative or positive. When the slope is negative, r is negative.

When the slope is positive, r is positive.

When r is 1 or -1 , all the points fall exactly on the line of best fit:

When r is greater than .5 or less than $-.5$, the points are close to the line of best fit:

When r is between 0 and .3 or between 0 and $-.3$, the points are far from the line of best fit:

When r is 0, a line of best fit is not helpful in describing the relationship between the variables:

Example

Two variable x and y

- $X = [1, 2, 3, 4, 5]$ (e.g., number of hours studied)
- $Y = [50, 55, 60, 65, 70]$ (e.g., corresponding test scores)

We want to calculate the correlation coefficient r using the formula:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Step 1: Calculate the means of X and Y

$$\bar{X} = \frac{2 + 4 + 6 + 8 + 10}{5} = \frac{30}{5} = 6$$
$$\bar{Y} = \frac{50 + 55 + 60 + 65 + 70}{5} = \frac{300}{5} = 60$$

Step 2: Calculate the deviations from the mean

For X :

$$X_i - \bar{X} = [2 - 6, 4 - 6, 6 - 6, 8 - 6, 10 - 6] = [-4, -2, 0, 2, 4]$$

For Y :

$$Y_i - \bar{Y} = [50 - 60, 55 - 60, 60 - 60, 65 - 60, 70 - 60] = [-10, -5, 0, 5, 10]$$

Step 3: Calculate the product of deviations

$$(X_i - \bar{X})(Y_i - \bar{Y}) = [-4 \cdot -10, -2 \cdot -5, 0 \cdot 0, 2 \cdot 5, 4 \cdot 10] = [40, 10, 0, 10, 40]$$

$$\sum (X_i - \bar{X})(Y_i - \bar{Y}) = 40 + 10 + 0 + 10 + 40 = 100$$

Step 4: Calculate the squares of deviations

For X :

$$(X_i - \bar{X})^2 = [-4^2, -2^2, 0^2, 2^2, 4^2] = [16, 4, 0, 4, 16]$$

$$\sum (X_i - \bar{X})^2 = 16 + 4 + 0 + 4 + 16 = 40$$

For Y :

$$(Y_i - \bar{Y})^2 = [-10^2, -5^2, 0^2, 5^2, 10^2] = [100, 25, 0, 25, 100]$$

$$\sum (Y_i - \bar{Y})^2 = 100 + 25 + 0 + 25 + 100 = 250$$

Step 5: Apply the formula

$$r = \frac{100}{\sqrt{40 \cdot 250}} = \frac{100}{\sqrt{10000}} = \frac{100}{100} = 1$$

```
import pandas as pd

data = {
    "Study Hours (X1)": [2, 4, 6, 8, 10],
    "Test Scores (X2)": [50, 55, 60, 65, 70]
}

df = pd.DataFrame(data)

correlation_matrix = df.corr()

print(correlation_matrix)
```